Paper 2147: Revision Plan in Simple Words

1. The One Big Change: Quantized Edge Ensemble

Today: both full-size models look at every image, we average their answers, and we say it runs on a Raspberry Pi without measuring anything.

After revision: we keep the same two-model ensemble, but use the **small DINOv2 (ViT-S)** and convert **both models to 8-bit (INT8)**, so they are smaller and faster. Then we run everything on the Pi and measure time per image, RAM and model size.

Why we are doing it:

- Reviewer 1 said "edge-deployable" needs proof. The 8-bit models make the Pi version practical, and we measure it.
- Reviewer 3 asked how DINOv2 can run on small devices. Answer: in a smaller, 8-bit version, with measured speed and memory.
- Reviewer 1 said our contribution is small. The new contribution is a measured, quantized edge ensemble with a melanoma safety rule and calibrated confidence, tested on a proper held-out test set.
- Our biggest mistake is melanoma called Nevus. The melanoma safety rule predicts melanoma whenever its probability crosses a threshold, even if it is not the top class.

Honest note: the Pi runs both models on every image, so time per image is the sum of both. Quantization is what keeps that time reasonable, so the Pi numbers carry the edge claim.

2. Reviewer 1

Reviewer said	What it means	What we will do
Abstract shows validation accuracy; show test results.	We scored on the same data we tuned on.	Keep 15% of images aside, use them only once at the end, and report those scores.
Ensemble is a modest contribution.	Two fixed weights is not new enough.	Add the quantized edge ensemble, melanoma safety rule and calibration (Section 1).
Edge claim needs latency and memory on the device.	We said "edge" but measured nothing.	Run full-size and 8-bit versions of each model and the ensemble on the Raspberry Pi; record time per image, RAM and model size.
Weights tuned on validation must be tested separately.	Same as the first point.	Choose ensemble weights and the melanoma threshold on validation, lock them, then run the test set once.
Compare with published ISIC 2019 results.	Show where we stand against others.	Add a table of other papers' results, noting where their setup differs from ours.

Reviewer said	What it means	What we will do
Conclusion should discuss the small 1.24-point gain.	Be honest that the gain was small.	Say so plainly, and explain that the main benefit of the new version is a measured, practical Pi deployment with better melanoma recall.

3. Reviewer 2

Reviewer said	What it means	What we will do
Only three classes; justify why.	Explain the choice.	They are the three most common ISIC 2019 classes, and melanoma vs Nevus is the most dangerous confusion.
Report per-class recall, especially melanoma.	Missing melanoma matters most.	Show recall for each class, with melanoma first. Add the melanoma safety rule.
Describe class balance.	How many images per class.	Table of image counts per class for train, validation and test.
Confirm lesion-level splitting.	Photos of one lesion must not be in both train and test.	Split the data again by lesion ID so each lesion is in only one set.
Test statistical significance.	Prove the gain is not luck.	Run McNemar's test and give confidence ranges for the scores.
State limits of the three-class subset.	Admit what we did not cover.	Write a short limitations paragraph.

4. Reviewer 3

Reviewer said	What it means	What we will do
Discuss DINOv2 on edge devices.	The big model may be too heavy for a Pi.	Test the small DINOv2 in 8-bit form on the Pi and compare it with the big one.
Grad-CAM visualisations would help.	Show where the model looks.	Add heatmaps for correct cases and for melanoma-called-Nevus mistakes.
Report training settings.	Others must be able to repeat our work.	One table with every setting: learning rate, epochs, batch size, optimizer, augmentation.
Discuss fairness across skin tones.	Does it work on darker skin?	Estimate skin tone from image colour and check melanoma recall per group. Note the dataset has few dark-skin images.
Cite recent edge-oriented classifiers.	Show we know current work.	Add 4 to 6 recent papers on skin models for phones or small devices.
Future work could include all ISIC classes.	Mention the next step.	Say we will extend to all 8 classes and test on other datasets.

5. Work Order (9 Days)

Days	Work
1 to 2	New split by lesion; retrain both models.
3	Make 8-bit versions of the models.

Days	Work
4 to 5	Pi measurements for full and 8-bit models (Chetan); choose ensemble weights and melanoma threshold on validation.
6	One final test run; significance test; heatmaps; skin-tone check; all new graphs.
7 to 8	Write the paper fresh in the Procedia template.
9	Proofread, similarity report, response letter.

All graphs and numbers come out new automatically, because the data split and models are new.

6. What Changes in the Paper

Title, abstract and introduction

Part	Now (submitted)	After revision
Title	Edge-Deployable Skin Lesion Classification: A Weighted Ensemble of MobileNetV2 and DINOv2	Edge-Deployable Skin Lesion Classification with a Quantized CNN-Transformer Ensemble (see Section 7)
Abstract	Validation accuracy of both models, fixed 0.4 / 0.6 weights, 91.36% ensemble accuracy, Raspberry Pi mentioned without numbers.	Quantized edge ensemble in one line, test-set accuracy, melanoma recall, Pi time per image and RAM, accuracy kept after 8-bit conversion, significance result. All numbers new.

Part	Now (submitted)	After revision
Introduction: problem	Lesions look alike; combine a CNN and a transformer.	A CNN-transformer ensemble is accurate but heavy for a Pi; it must be made small enough to run there, and missing melanoma is the costliest error.
Introduction: earlier work	Not mentioned.	One sentence citing the ICAICCIT paper as the earlier full-size ensemble that this work extends to the edge.
Introduction: contri-butions	Train two models, combine with fixed weights, build Streamlit app.	Quantized edge ensemble with small DINOv2, melanoma safety rule, calibrated confidence, lesion-level held-out test, on-device Pi measurements, significance and explainability analysis.

The two epoch graphs (training curves)

The submitted paper has two: MobileNetV2 accuracy and loss over 50 epochs, and DINOv2 accuracy and loss over 50 epochs. They will be **regenerated, not edited**:

- **Why they change:** the curves come from training on a data split. We are making a new lesion-level split, so both models are retrained and every point on the curves is new.

- **DINOv2 graph:** it will show the small DINOv2 (ViT-S), which is the version used on the Pi, so it is a different model as well as different data.
- **Shape will look familiar:** accuracy going up and loss going down is how every training curve looks. That is normal; the values, the best epoch and the gap between training and validation will differ.
- **Never reuse or restyle the old graphs.** IEEE owns the published figures, and new graphs come for free from the new runs.
- **Layout suggestion:** put both models in one figure with four panels (accuracy and loss for each) to save space for the new graphs.

Graphs we are adding

New graph	What it shows	Answers
Accuracy vs Pi time	Each model and the ensemble, full-size and 8-bit, as one dot: how accurate and how fast.	Reviewer 1: edge latency; Reviewer 3: DINOv2 on edge
Pi measurements bar chart	Time per image, RAM and model size, full-size vs 8-bit.	Reviewer 1: latency and memory; Reviewer 3: DINOv2 on edge
Accuracy before and after 8-bit	How much accuracy is lost by quantization (should be very small).	Reviewer 3: DINOv2 on edge; Reviewer 1: modest contribution
New confusion matrices	Mistakes per class on the new test set, without and with the melanoma safety rule.	Reviewer 2: per-class and melanoma recall

New graph	What it shows	Answers
Reliability diagram	Whether the confidence score shown in the app can be trusted, before and after calibration.	Reviewer 1: test results; supports the app's confidence score
Heatmaps (Grad-CAM)	Where each model looks, for correct cases and melanoma-called-Nevus mistakes.	Reviewer 3: Grad-CAM
Skin-tone recall chart	Melanoma recall for each skin-tone group.	Reviewer 3: fairness across skin tones

Graphs replaced: the old block diagram is redrawn to show the quantized models running on the Pi, and the old sample images are replaced with images from the new test set.

7. Title Suggestions

#	Title
1	**Edge-Deployable Skin Lesion Classification with a Quantized CNN-Transformer Ensemble (recommended)**
2	Quantized CNN-Transformer Ensemble for Melanoma-Sensitive Skin Lesion Classification on Raspberry Pi
3	Calibrated and Quantized CNN-Transformer Ensemble for On-Device Dermoscopic Classification

Go with title 1: it keeps "Edge-Deployable" from the submitted title so the editor sees the same paper improved, names what is new (quantization), and uses no model names.